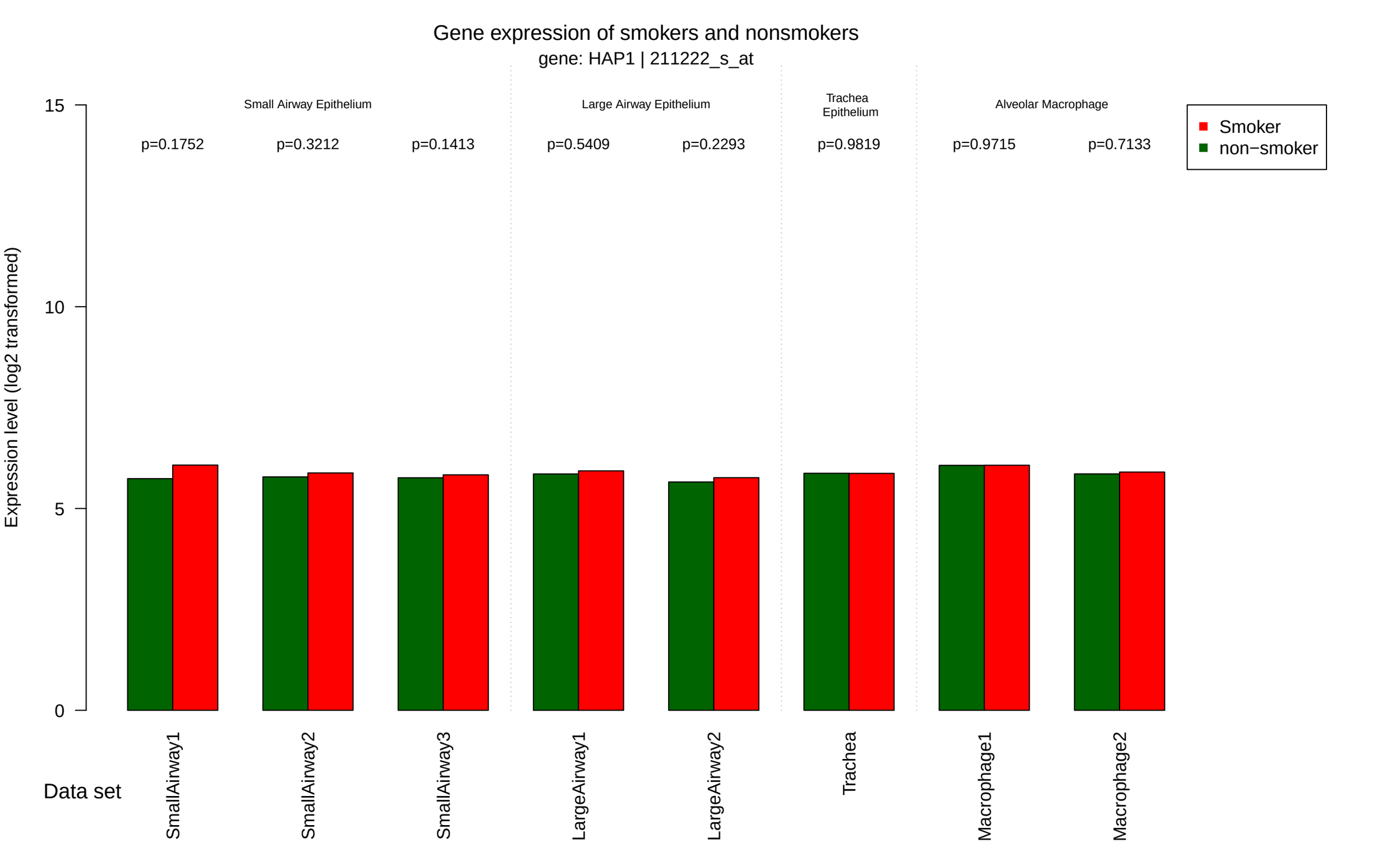
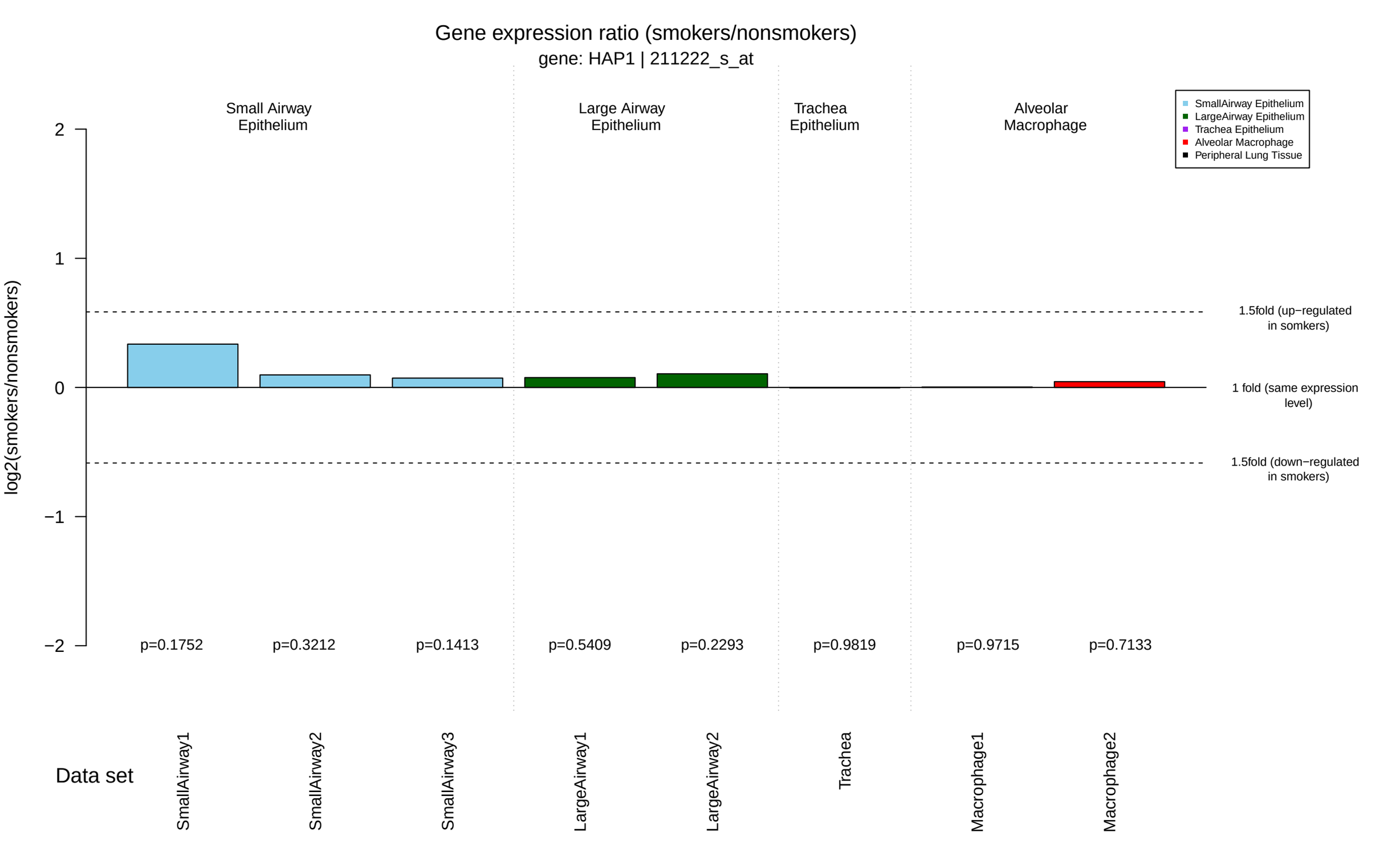
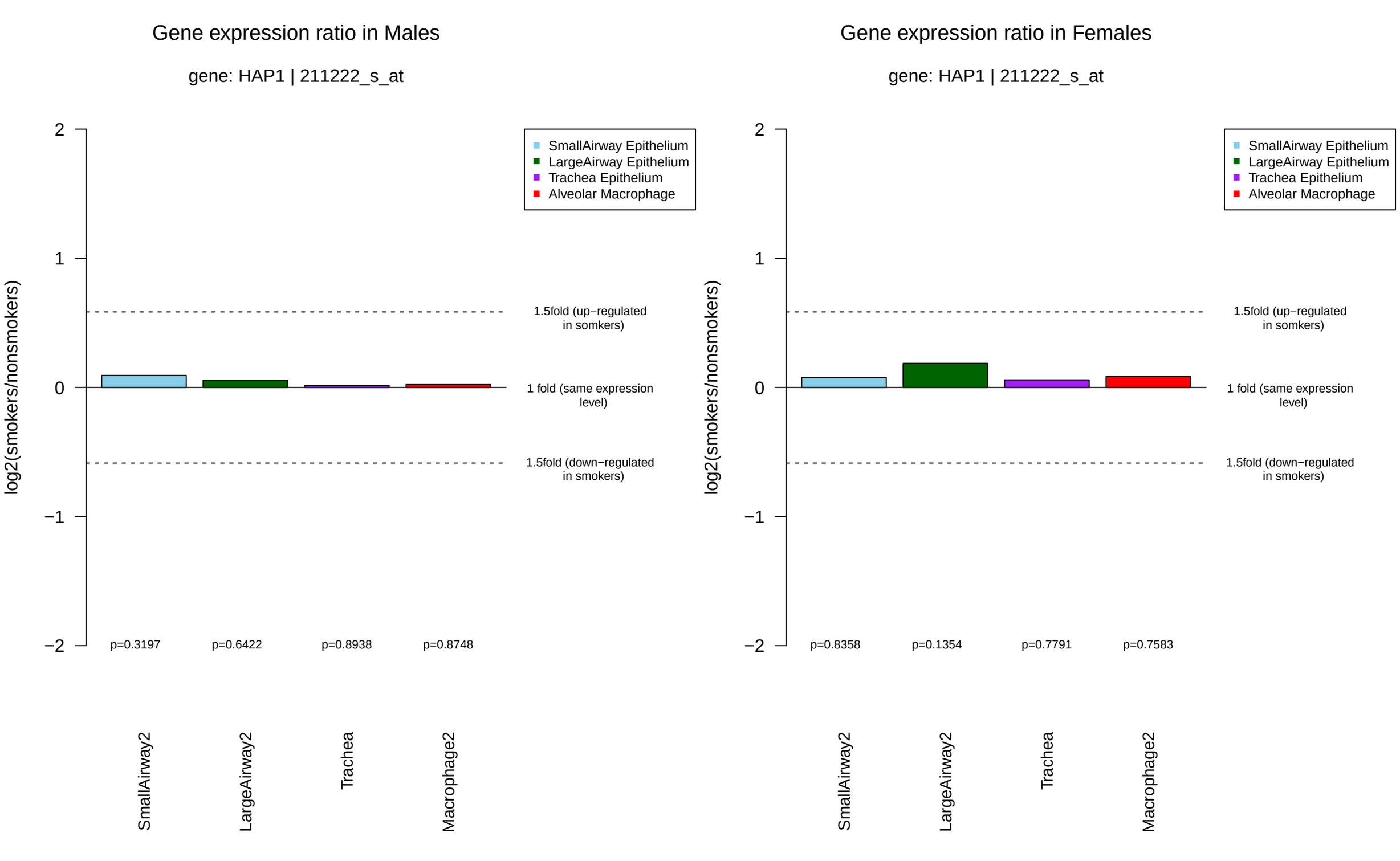
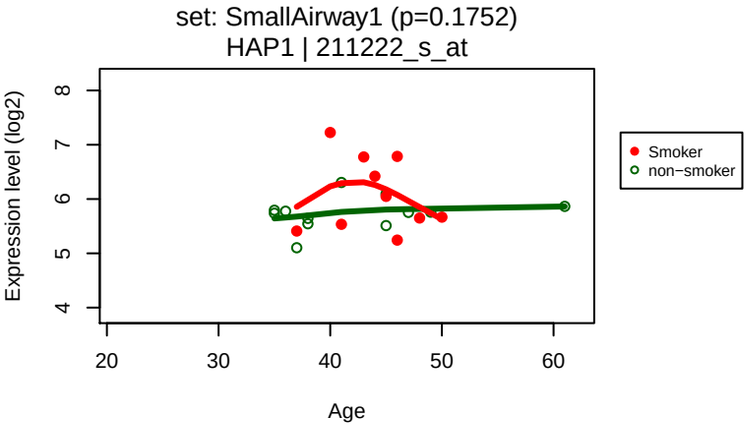
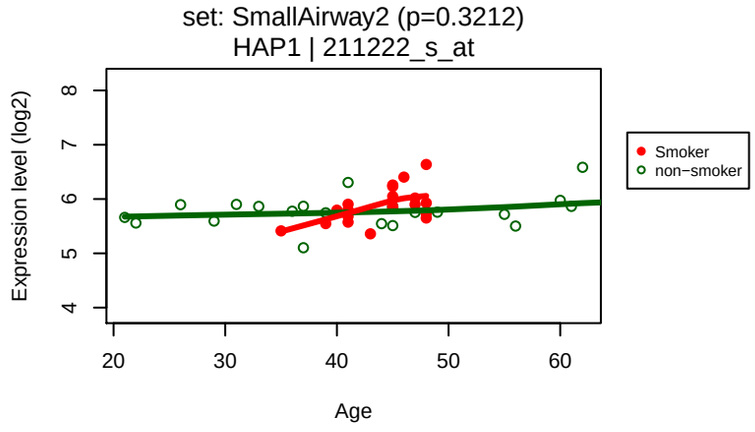
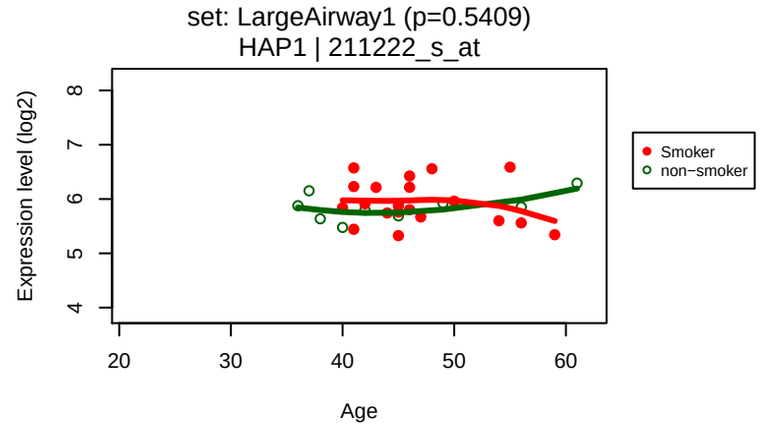

Gene expression of smokers and nonsmokers
gene: HAP1 | 211222_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1752
p=0.3212
p=0.1413
p=0.5409
p=0.2293
p=0.9819
p=0.9715
p=0.7133
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

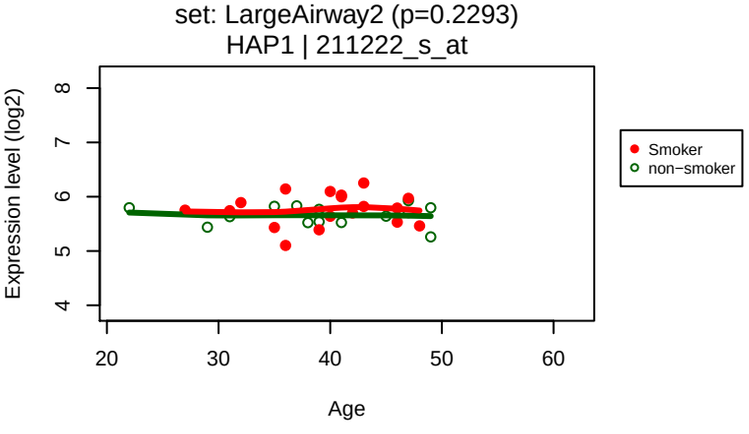

Gene expression ratio (smokers/nonsmokers)
gene: HAP1 | 211222_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1752
p=0.3212
p=0.1413
p=0.5409
p=0.2293
p=0.9819
p=0.9715
p=0.7133
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

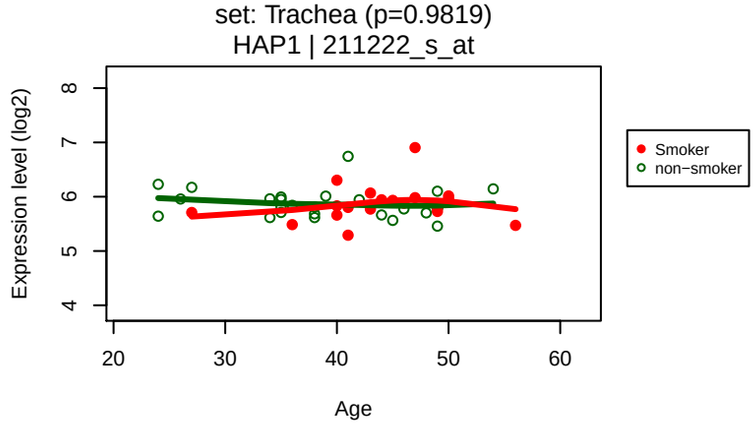

Gene expression ratio in Males
gene: HAP1 | 211222_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3197
p=0.6422
p=0.8938
p=0.8748
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HAP1 | 211222_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8358
p=0.1354
p=0.7791
p=0.7583
SmallAirway2
LargeAirway2
Trachea
Macrophage2

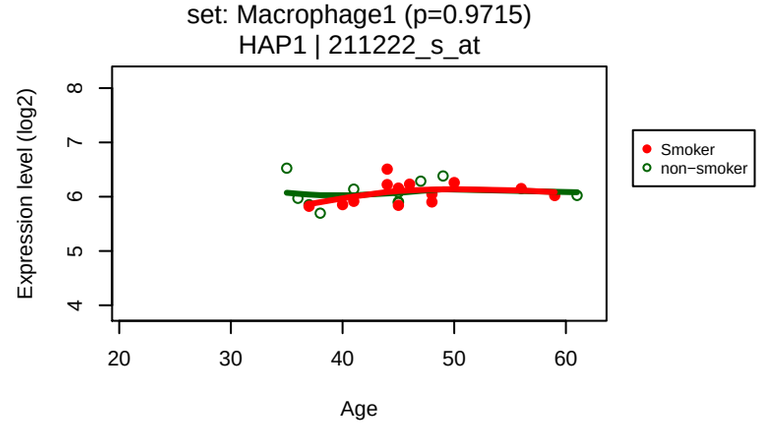

set: SmallAirway1 (p=0.1752)
HAP1 | 211222_s_at
Expression level (log2)
Age
Smoker
non−smoker

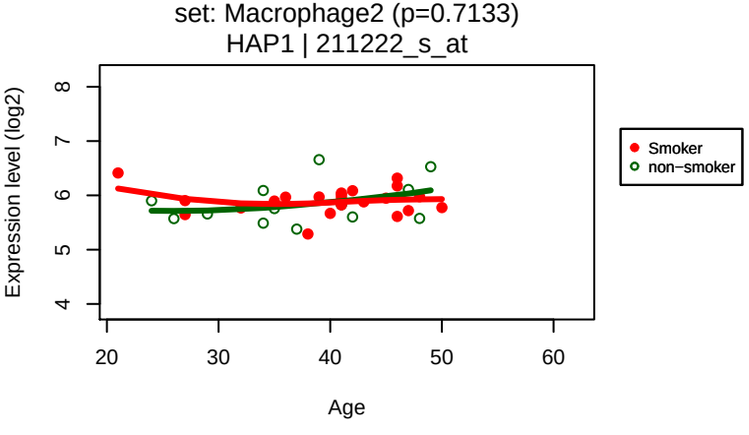

set: SmallAirway2 (p=0.3212)
HAP1 | 211222_s_at
Expression level (log2)
Age
Smoker
non−smoker

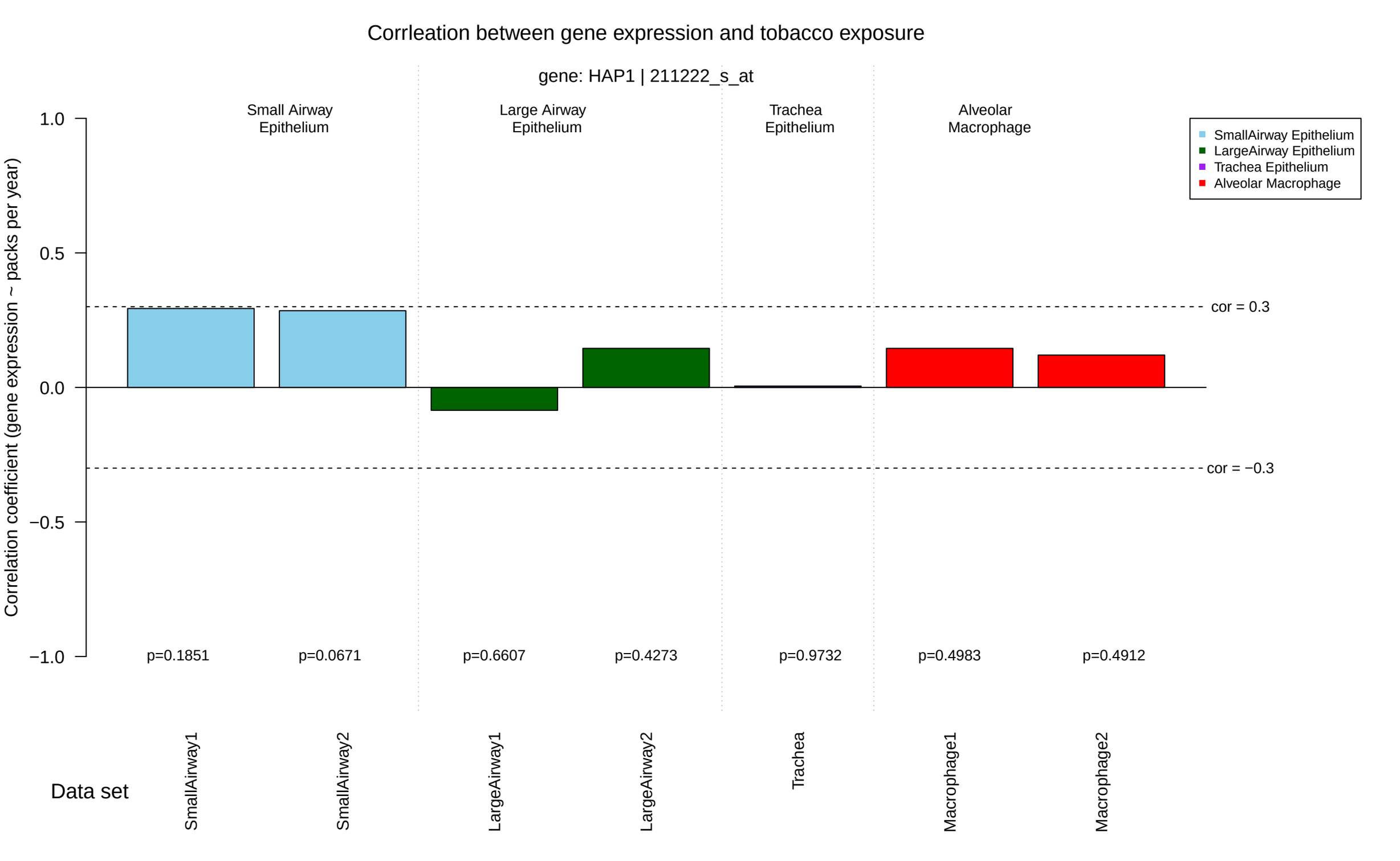

set: LargeAirway1 (p=0.5409)
HAP1 | 211222_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2293)
HAP1 | 211222_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9819)
HAP1 | 211222_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9715)
HAP1 | 211222_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7133)
HAP1 | 211222_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HAP1 | 211222_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.1851
p=0.0671
p=0.6607
p=0.4273
p=0.9732
p=0.4983
p=0.4912
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2